Network Gateway System

High-Level Design (HLD)

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This document describes the software architecture of the central network gateway: CAN routing, Ethernet backbone bridging, diagnostic multiplexing and time synchronization components and interfaces.

1 Revision history

Revision	Date	Author	Description
A	2026-05-14	A. Kumar	Initial architecture baseline

2 Scope and context

2.1 Scope

The central gateway covers inter-bus routing between chassis CAN, body CAN, powertrain CAN and the Ethernet backbone, plus diagnostic multiplexing. Out of scope: physical transceiver design and production connector pin assignment.

2.2 System context

The gateway runs on GatewayEcu and interconnects ChassisEcu, PowertrainEcu, BodyEcu and the ADAS domain over a switched Ethernet backbone.

3 Software architecture

3.1 Component breakdown

Component	Type	Description
CanRoutingTableManager	ApplicationSwComponentType	CAN frame routing tables
EthernetBridgeController	ApplicationSwComponentType	Ethernet backbone bridging
DiagnosticMultiplexer	ApplicationSwComponentType	UDS diagnostic routing
TimeSyncCoordinator	ApplicationSwComponentType	Global time synchronization
SecurityGateway	ApplicationSwComponentType	Secure onboard communication
GatewaySignalService	ServiceSwComponentType	Gateway signal service
CentralGatewayAbstraction	EcuAbstractionSwComponentType	Gateway hardware abstraction
NetworkTopologyComposition	CompositionSwComponentType	Composition of gateway components

3.2 Ports and interfaces

Port	Direction	Interface	Component
CanRoutingIn	required	CanFrameI	CanRoutingTableManager
EthernetBridgeOut	provided	EthernetFrameI	EthernetBridgeController
DiagRouteOut	provided	DiagRoutingI	DiagnosticMultiplexer
TimeSyncOut	provided	GlobalTimeI	TimeSyncCoordinator
SecuredFrameOut	provided	SecuredFrameI	SecurityGateway
GatewayServiceIn	required	GatewayServiceI	GatewaySignalService
TopologyQueryOut	provided	TopologyQueryI	CentralGatewayAbstraction
StatusReportOut	provided	GatewayStatusI	SecurityGateway

3.3 Signal inventory

Signal	Interface	Sender	Receiver	Bits
ChassisCanFrame	CanFrameI	CanRoutingTableManager	EthernetBridgeControl	64
BackboneFrame	EthernetFrameI	EthernetBridgeControlle	DiagnosticMultiplexer	128
UdsResponse	DiagRoutingI	DiagnosticMultiplexer	CanRoutingTableManage	32
GlobalTimeValue	GlobalTimeI	TimeSyncCoordinator	SecurityGateway	32
SecuredPayload	SecuredFrameI	SecurityGateway	EthernetBridgeControl	96
GatewayMode	GatewayModeI	GatewaySignalService	CanRoutingTableManage	4
TopologyVersion	TopologyQueryI	CentralGatewayAbstracti	GatewaySignalService	8

4 Functional flows

4.1 Chassis to backbone routing flow

CanRoutingTableManager receives ChassisCanFrame from the chassis CAN bus and forwards converted frames to EthernetBridgeController for backbone transmission. SecurityGateway signs the payload before transmission through SecuredFrameI.

4.2 Diagnostic multiplexing flow

DiagnosticMultiplexer routes UDS requests from the Ethernet backbone to the addressed ECU over the corresponding CAN bus and returns UdsResponse to the tester. CanRoutingTableManager forwards the response frames.

4.3 Time synchronization flow

TimeSyncCoordinator broadcasts GlobalTimeValue to all connected ECUs. SecurityGateway rejects frames with stale timestamps to protect against replay attacks.

5 Timing and runnables

TimingEvent RoutingFlushEvent triggers FlushRoutingTablesRunnable every 10 ms. TimingEvent TimeSyncEvent triggers BroadcastGlobalTimeRunnable every 100 ms. TimingEvent SecurityCheckEvent triggers VerifyFrameSecurityRunnable every 10 ms. All runnables are scheduled by GatewayTask on GatewayEcu.

6 Basic software integration

Signal routing uses Com and PduR across all CAN buses. Incoming Ethernet frames are handled by SoAd and TcpIp. NvM stores the routing tables and security keys. Rte provides the port API used by all gateway components.

Mode management is performed by BswM and EcuM. Watchdog supervision for GatewayTask is performed by WdgM. The Ethernet switch configuration is applied through Spi during startup.

The diagnostic path routes UDS requests through Dcm and Dem. GatewaySignalService consumes GatewayServiceI and forwards topology changes to the network management stack.

7 Diagnostics and configuration

DTCs for routing table checksum mismatch, time sync loss and security verification failure are defined in the diagnostics specification. The HLD only defines the reporting path: component to Dcm to CanIf.

Configuration parameters are grouped into the domain configuration container and written through NvM during shutdown.

8 Known limitations

Routing tables are static in revision A; dynamic reconfiguration requires secure OTA support not covered here.

9 Open questions

Open: calibration ownership for cross-domain interfaces and the timeline for production software freeze are tracked outside this document.